GE1501 Cornerstone of Engineering I

Assessment Brief: AE1 Robot Project

Assessment Details

Course Leader:	Dr Sophia Economides
Issue Date:	14 September 2026
Assessment Type:	Project
Assessment Title:	AE1 Robot Project – Group and Individual Work
Assessment Weighting:	60% of the course
Hand-in Deadline:	Presentations and live demonstrations take place during class on 8, 9 or 11 December, depending on your section. Slides, video and code must be submitted on the same day as your presentation, at 23:00. The individual report is due on 11 December at 23:00 for all sections.
Planned Feedback Deadline:	28 calendar days after hand-in deadline or last presentation date
File Format Accepted:	Video: mp4, embedded via Panopto. Code: py, pdf, docx or txt. Report: pdf or docx. Slides: pptx or pdf.
Mode of Submission:	Online (Canvas)
Anonymous Submission:	The slides, code and video are team submissions and are not anonymous. The individual report is submitted and marked anonymously.
Marking Scheme:	Numerical

✓	**Permitted**	Use of AI is permitted on this assignment. Specific tools and uses must be declared using the AI Use Declaration format (example: Claude used for feedback on a draft). See the AI Use in Assignments page for full guidance.

Project Overview

In this project you will build and programme a robot, working in a team of two. Your robot must navigate a corridor provided in the makerspace, using ultrasonic sensors to detect and avoid the walls. At the end of the corridor, a random signal will select one of three tasks for your robot to perform:

1. sounding a buzzer
2. activating a light, or
3. completing a 360 degree rotation.

You will not know in advance which task will be selected, so your robot must be programmed and tested to perform all three.

On the day of your presentation, you will demonstrate your robot live. If your robot cannot perform, you may instead show a previously recorded video of a successful attempt. You will also present your slides, answer questions and have your robot's build quality checked. All of this takes place within a ten minute slot.

Every team member must submit a copy of the slides and video on the day of the presentation. All team members receive the same mark for these components. You also receive an individual mark for your presentation and for how you answer questions.

Separately from this, you must submit an individual report describing your work.

Project Tasks

The project has three components:

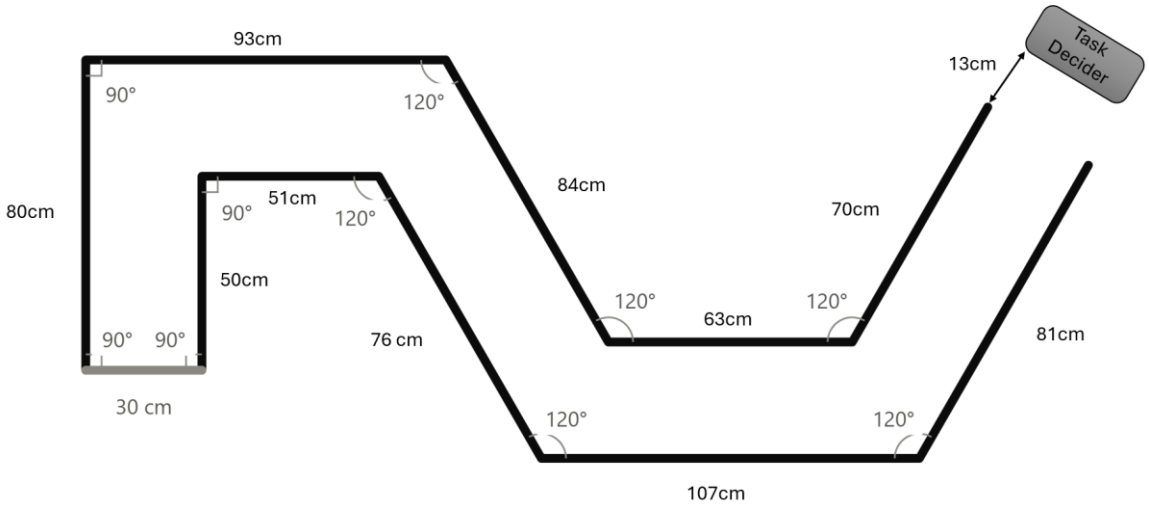
- Course Navigation and Task Completion (40%, group work)
- Individual Engineering Report (40%, individual work)
- Group Presentation (20%, group and individual)

Task 1: Course Navigation and Task Completion (40%, Group Work)

This task is assessed on four things: how well your robot navigates the corridor, how well it detects and carries out the selected task, the quality of your code and the build quality of your robot.

Course Navigation

Build and programme your robot to navigate the course below autonomously, using ultrasonic sensors to detect and avoid the walls. The course layout is shown below. All dimensions are in centimetres.



Task Detection and Execution

At the end of the corridor, a random signal will select one of three tasks for your robot to perform: sounding a buzzer, activating a light, or completing a 360 degree rotation. Your robot must correctly detect the signal and carry out the selected task well.

Code Quality

You must submit code covering all three possible tasks. Your code should be well organised and work together as one coherent programme, rather than as separate, disconnected pieces.

Technical Merit and Build Quality

Build your robot neatly and securely: tidy wiring, securely mounted components, and no loose or exposed connections. It should hold up to normal handling.

Submission Requirements for Task 1

- One video, uploaded to Canvas via Panopto, showing your course navigation and task completion. Include timestamps marking the start of the course navigation and the start of the task demonstration. See the separate Video Recording and Submission Guide for how to prepare and upload this.
- One code submission (a single file or a clearly organised folder) containing the navigation code, the task detection code and all three task functions, clearly sectioned or commented so each part can be identified and marked separately. Attach a completed AI Use Declaration table, or state that AI was not used.

Each team member submits their own identical copy of the video and code. These are not anonymous. The mark awarded to one team's submission applies to every member of that team.

A fixed deduction applies to the Course Navigation and Task Completion marks if the video is missing, cannot be opened, or is of such poor quality that the task cannot be clearly identified as working.

Task 2: Individual Engineering Report (40%, Individual Work)

The report is an opportunity to practise your engineering documentation skills, begin compiling a record of engineering projects for co-op and job applications and reflect on the engineering design process.

Report Requirements

- Word limit: maximum 1,500 words (excluding list of references, captions, headings etc.). Up to 10% over or under is acceptable. Anything beyond 10% over will not be marked.
- Format: PDF or Word file only. No shared links.

The report must include:

- A clear explanation of how you applied the engineering design process to this project
- An explanation of how your robot works, including its navigation, task detection and task execution
- Show the steps you followed, including the trials and tests you carried out and what you learned from them. See the Report Guidance page on Canvas for more detail.
- A discussion of your approach and reasoning, meaning the specific decisions your group made within the constraints of the kit, for example sensor placement, code structure or how you approached calibration, rather than a general design comparison
- Photographs from your testing and prototyping work
- A short section on your individual contribution, describing the specific tasks you personally carried out and what you learned from them. This should describe your own role, not the team's work as a whole.
- References in Harvard format, with in-text citations, for any technical background, component specifications or external sources used

Different parts of the report must be clearly indicated, following the format set out in the Report Guidance page on Canvas. Attach a completed AI Use Declaration table as an appendix, or state that AI was not used.

See the Report Guidance page on Canvas for advice on structuring and writing your report. Questions can be raised during class, office hours, by email or via Canvas.

Task 3: Group Presentation (20%, Group and Individual)

Present your project results and design process to the class as a team, with the help of slides.

Your presentation must include as a minimum the five slides set out in the Presentation and Slide Guidance page on Canvas:

- title,
- approach and reasoning,
- what worked and what did not,
- results, and
- conclusion.

See that page for the full slide structure, slide design guidance and the referencing format to use on your slides.

Timing

Your presentation, including your live demonstration and any setup or take down time, must last no longer than 10 minutes. Presentations running over will be stopped, with a grade reduction applied. Plan your presentation time so each team member has a genuine opportunity to contribute.

Marking Breakdown

- Slide Content: 25%
- Slide Quality and Appearance: 10%
- Citations: 5%
- Timing: 10%
- Individual Mark: 50%

Slide Content, Slide Quality and Appearance, Citations and Timing are marked as teamwork. Individual Mark is assessed separately for each student.

The Individual Mark

The Individual Mark is based on how clearly you communicate the content on your own assigned slides, and on how well you answer questions about the project during both the presentation question period and the live demonstration. Any team member may be asked about any part of the project, not only the part they personally worked on.

What to Submit

Each team member submits their own identical copy of the slides on the same day as the presentation. These are not anonymous. The marks awarded to one team's submission apply to every member of that team. Attach a completed AI Use Declaration table as an additional final slide, or state that AI was not used. This declaration slide is in addition to, and does not count towards, the required slide structure above. It does not need to be presented at the live presentation.

Assessment Criteria

The rubrics for each component are set out below.

Course Navigation and Task Completion Rubric (out of 100)

Component	Excellent	Good	Adequate	Weak	Not achieved
Course Navigation (40 points)	36-40: No wall contact, one continuous run with no pauses to reorient, stays centred throughout	30-35: No wall contact, reasonably centred, but with a pause or hesitation at one or more points, for example slowing before a turn	22-29: One or two light wall contacts that do not stop the robot, or drifts to one side rather than holding a central line, but completes unaided	10-21: Repeated or hard wall contact, or consistently hugs one side of the path, or needs a nudge or restart to continue	0-9: Does not complete autonomously, or needs more than one manual intervention to finish
Task Detection and Execution (20 points)	18-20: Immediate, correct detection, task executed cleanly, for example a prompt clear buzzer, a light that activates properly, or a complete smooth rotation	14-17: Correct detection with at most a brief delay, task executed correctly but with a minor flaw, for example a short hesitation before the buzzer, or a rotation that slightly overshoots	10-13: Detection is correct but slow, or the task is recognisable but flawed, for example a faint or intermittent buzzer, or a rotation that stalls before continuing	5-9: Detection is initially incorrect and needs a retry, or the task is only partially recognisable as correct	0-4: No working detection, the wrong task is performed, or the task is not attempted
Code Quality (30 points)	27-30: All three task functions correctly implemented, efficient and well organised, for example a clean decision structure selecting between tasks, no redundancy	22-26: All three functions work correctly, though somewhat repetitive or less efficiently organised	17-21: Functions mostly work, but the code is poorly organised, with redundancy or inconsistency affecting readability	8-16: One or more functions do not work correctly, or the code is fragmented with little structure	0-7: Code is missing, non-functional, or shows no coherent structure
Technical Merit and Build Quality (10 points)	9-10: Neat, secure, robust construction throughout	7-8: Generally tidy, with a minor issue such as a loose wire or component	5-6: Functional but visibly untidy	2-4: Loose, insecure or fragile construction affecting reliability	0-1: No evidence of care in construction

Individual Engineering Report Rubric (out of 100)

Component	Excellent	Good	Adequate	Weak	Not achieved
Title and Abstract (10 points)	9-10: Clear title, abstract concisely summarises the whole project	7-8: Minor clarity issues in title or abstract	5-6: Adequate but underdeveloped	2-4: Poorly separated from the main text or unclear	0-1: Title or abstract missing
Design (35 points)	32-35: All elements present, well organised, logical flow. Trials and tests are clearly documented, showing what was tried, what happened, and what changed as a result	25-31: Most elements present and well organised. Trials are documented, but with some lack of clarity	18-24: Elements present but thin. Trials are mentioned in general terms rather than documented clearly	7-17: Poor organisation. Little evidence of systematic testing	0-6: Not submitted or no effort shown
Individual Contribution (10 points)	9-10: Specific, credible account of the student's own role and tasks, distinct from the team's work as a whole	7-8: Mostly specific, minor generic statements	5-6: Some specific detail but largely generic	2-4: Vague or largely indistinguishable from team description	0-1: Not included
Summary and Conclusion (25 points)	23-25: Clear description of final design, thoughtful reflection on lessons and future application	18-22: Solid but less detailed	13-17: Basic description and limited reflection	5-12: Weak or incomplete	0-4: Not submitted
References (10 points)	9-10: Complete, correctly formatted, all in-text citations matched, minimum 3 sources	7-8: Minor formatting errors	5-6: Incomplete list or format errors	2-4: Very incomplete or incorrect format	0-1: No references included
English Proficiency (10 points)	9-10: Excellent technical writing, minimal errors	7-8: Good, few errors	5-6: Adequate, some errors not impeding understanding	2-4: Multiple errors impeding communication	0-1: Cannot assess language proficiency

Group Presentation Rubric (out of 100)

Component	Excellent	Good	Adequate	Weak	Not achieved
Slide Content (25 points)	23-25: All required slides present and fully developed, clear and specific content throughout	18-22: All required slides present, well developed, with one slide noticeably thinner	13-17: Required slides present but generic or underdeveloped in places	6-12: Several slides missing content or structure	0-5: Slides missing or not attempted
Slide Quality and Appearance (10 points)	9-10: Professional, consistent design, clear images, appropriate text density, slides numbered	7-8: Minor inconsistencies that do not affect clarity	5-6: Noticeable issues, for example inconsistent formatting or unclear images	2-4: Cluttered or inconsistent throughout	0-1: No slides or unusable
Citations (5 points)	5: Correct citation format used consistently wherever a source is referenced	4: Minor formatting inconsistencies	3: Present but inconsistent or incomplete	1-2: Largely missing or incorrect where expected	0: No citations where sources are clearly used
Timing (10 points)	9-10: Finishes within the slot with time to spare, time clearly and fairly divided so every member has a genuine opportunity to contribute	7-8: Finishes within the slot, division mostly fair with a minor imbalance	5-6: Finishes within the slot, but one member noticeably dominates	2-4: Runs over and is stopped, or division is clearly unfair	0-1: Significantly over time, or one member barely contributes
Individual Mark (50 points)	45-50: Communicates their own content clearly and accurately, answers questions about any part of the project well, with genuine engagement throughout	35-44: Communicates their own content well, with minor gaps in depth or occasional drift from the question	25-34: Communicates their own content reasonably, with some inaccuracy or vagueness in answers	10-24: Struggles to communicate their own content, for example mostly reading from the slide, with frequent inaccuracy or visible disengagement	0-9: Unable to meaningfully communicate their own content or answer questions, or clearly disengaged throughout

Submitting Assessments

Each student submits their own copy of the team components (video, code and slides). This way each student takes responsibility for ensuring their own copy is uploaded correctly and on time.

> *The video, code and slides are not anonymous, since team membership is visible on Canvas.*

The individual report is submitted separately by each student, anonymously, through a standard individual assignment.

All students must submit the entire group work submission as well as their individual work. The submitted group work must be identical between all group members.

You can always reach out to ask questions and support via email or messaging on Canvas, also during class and office hours. There are some sessions during the semester, based on the available time, dedicated to doing project work during class. While working on the robot during class hours, students have the opportunity to ask questions and solve problems and issues about their robot, presentation, videos and all submissions. You may seek support for technical troubleshooting, assignment requirement clarifications, presentation structure, video recording assistance, and written report formatting and referencing. Draft presentation materials may be reviewed for general structure and approach, but detailed content feedback will not be provided. No content specific advice will be given for written reports, including technical solutions, design analysis or engineering interpretations, to maintain assessment integrity. Early consultation is encouraged, particularly for those anticipating challenges with group work, presentation delivery or requiring accessibility support.

You have three submission attempts, but only the last submission will be graded. If your last submission attempt is late, you will receive the late penalty even if you have a previous submission that was on time. Please make sure to avoid multiple submissions for assessments with multiple components, as only the last attempt will be graded. Upload several files in one submission attempt instead.

If the assessment requires anonymous submission, please be sure you have left your name off your submission and out of the submission file name, as failing to do so may result in a 0% mark on the assessment.

Refer to the assessment details table for acceptable file formats. Avoid submitting zip files unless explicitly required by the assessment brief, use the add files function to submit multiple files instead. If you are submitting a physical artefact, you must also provide clear and thorough documentation, such as photographs or a video, of your submission by the deadline.

Please ensure that you tick the agreement box at the very bottom of your Canvas submission page, scroll down if you do not see it. This will enable you to select Submit Assessment. Please review the submitted file to ensure that everything is in order.

If you encounter any issues with submission, email a copy of your assignment before the deadline to the course leader copying student.assessments@nulondon.ac.uk along with screenshots of the problem on Canvas, showing a timestamp.

Marking

The University uses two assessment marking schemes, one for undergraduate and one for postgraduate, to mark all taught programmes leading to an award of the University. More detailed information on the assessment marking scheme and the criteria can be found in the Course Syllabus, available on the University's VLE.

Learning Outcomes

This assessment will enable students to demonstrate, in full or in part, the learning outcomes identified in the Course Descriptor. On successful completion of this assessment, students should be able to:

Knowledge and Understanding

- K1a: Demonstrate an understanding of the engineering design process
- K2a: Research the scientific principles and technical background required to understand the problem to be solved, and benchmark existing or related products
- K3a: Demonstrate an understanding of risk management in engineering practices

Subject Specific Skills

- S1a: Implement the desired design solutions using appropriate engineering techniques
- S2a: Analyse and interpret computational and experimental results as part of engineering practice
- S3a: Demonstrate the concept of engineering design by using modern drawing or sketching software tools and the principles of orthographic projection

Transferable Skills

- T1a: Communicate effectively with specialist and non-specialist audiences
- T2a: Apply problem solving skills to the conception of a particular solution

Accessing Feedback

Students can expect to receive feedback on all summative coursework within 28 calendar days of the submission deadline or, if applicable, the last oral assessment date, whichever is later. The 28 calendar day deadline does not apply to work submitted late. Feedback can be accessed through the assessment link on the Canvas course page.

Late Submissions

Please ensure that you submit your assignment well before the deadline to avoid any late penalties, as a submission made exactly on the deadline will be considered late. Please keep in mind that there may be differences between your computer's clock and the server time, which can cause discrepancies, and that Canvas may take some time to process your submission. All group members are responsible for the timely submission of the assessment.

Your Canvas submission portal displays two due dates, one is the deadline for your assignment, and the second is the latest possible date by which your assignment can be submitted late. Please make sure you submit by the assessment deadline in order to avoid late penalties.

If assessments are submitted late without approved Extenuating Circumstances, there are penalties. For assessments submitted up to one day late, any passing mark will receive 10 marks deducted or a threshold pass, 40% for undergraduate students, whichever is higher. Any mark below 40% for undergraduate students will stand. Students who do not submit their assessment within one day of the deadline, and have no approved Extenuating Circumstances, are deemed not to have submitted and to have failed that assessment element. The mark recorded will be 0%.

For further information, please refer to AQF7 Part C in the Academic Handbook.

Extenuating Circumstances

The University's Extenuating Circumstances procedure is in place if there are genuine circumstances that may prevent a student from submitting an assessment. If the EC application is successful, there will be no academic penalty for missing the published submission deadline.

Students are normally expected to apply for ECs in advance of the assessment deadline. Students may apply for consideration of ECs retrospectively if they can provide evidence that they could not have done so in advance of the deadline. All applications for ECs must be supported by independent evidence.

Students are reminded that the ECs procedure covers only short term issues, within 21 days leading to the submission deadline, and that if they experience longer term matters that impact on learning, they must contact Student Support and Development for advice.

Under the Extenuating Circumstances Policy, students may defer an assessed element on only one occasion and may request an extension on a maximum of two occasions.

For further information, please refer to the Extenuating Circumstances Policy in the Academic Handbook.

Academic Misconduct

Any submission must be a student's own work and, where facts or ideas have been used from other sources, these sources must be appropriately referenced. The University reserves the right to hold a viva if there are concerns about the authenticity of a student's or learner's work. The Academic Misconduct Policy includes the definitions of all practices that will be deemed to constitute academic misconduct. This includes the use of artificial intelligence where not expressly permitted within the assessment brief, or used in a manner other than specified, see the AI Use in Assignments page for the declaration requirements on this assessment.

Students should check the Academic Misconduct Policy before submitting their work. Students suspected of committing academic misconduct will face action under the Policy. Where students are found to have committed an offence, they will be subject to sanction, which may include failing an assessment, failing a course or being dismissed from the University, depending upon the severity of the offence committed. For further information, please refer to the Academic Misconduct Policy in the Academic Handbook.

Resubmission

For Task 1, students requiring resubmission (second or third sitting) must submit an updated video demonstrating resolved issues based on feedback provided, together with revised code showing the corrections made.
For Task 2, students requiring resubmission must identify the issues with their report and correct them based on feedback provided before resubmitting.
For Task 3, students requiring resubmission must submit a recorded presentation in a similar format to the original. If resubmission is required due to the Individual Mark rather than non-submission, the student will be asked further questions about the project to reassess their understanding. Other group members are not required to participate in an individual resubmission.